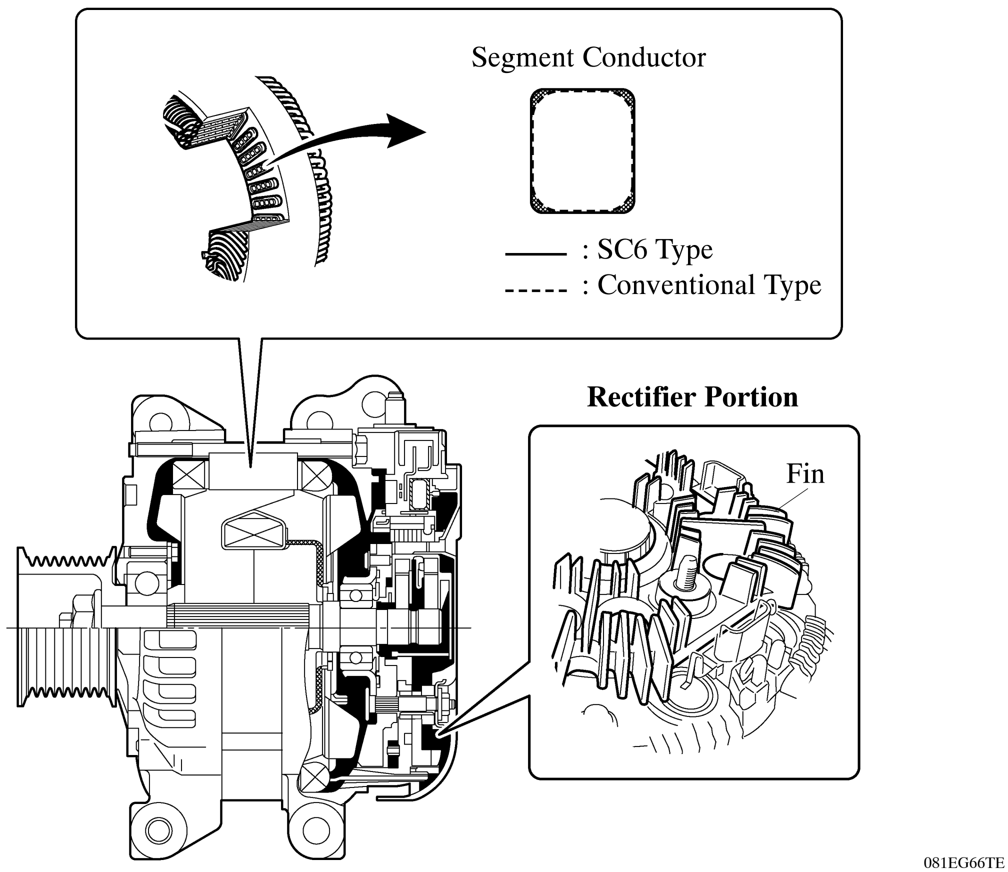
■CHARGING SYSTEM

1. General

- A compact and lightweight segment conductor type generator is used on all models.
- A SC6 type generator is provided on European models with the towing package. The cross sectional area of the segment conductors on a stator in this generator has been increased, which reduces the electrical resistance of the stator, generates high power output, and achieves a higher efficiency. This type uses an aluminum die-casting rectifier unit and also provides fins on the rectifier to improve heat dissipation.

Stator Portion



Generator Cross Section

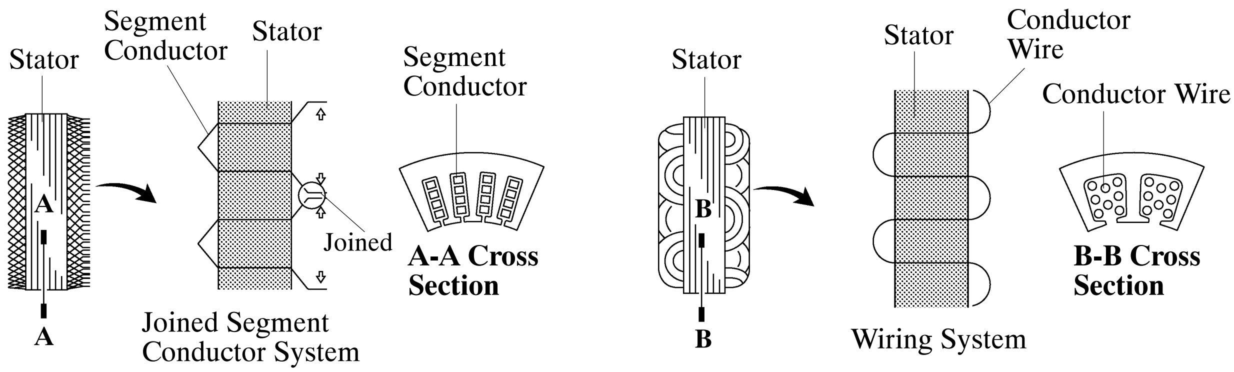
▶ Specifications ◀

Type	SC1	SC2	SC6
Destination	G.C.C. Countries	Europe*2	Europe*4
		Australia	
	General Countries*1	General Countries*3	
Rated Voltage	12V	←	←
Output Rated	130A	150A	180A

*1: Except for VX grade models
*2: European models without the towing package except cold area specifications
*3: Only on VX grade models
*4: Only on models with towing package

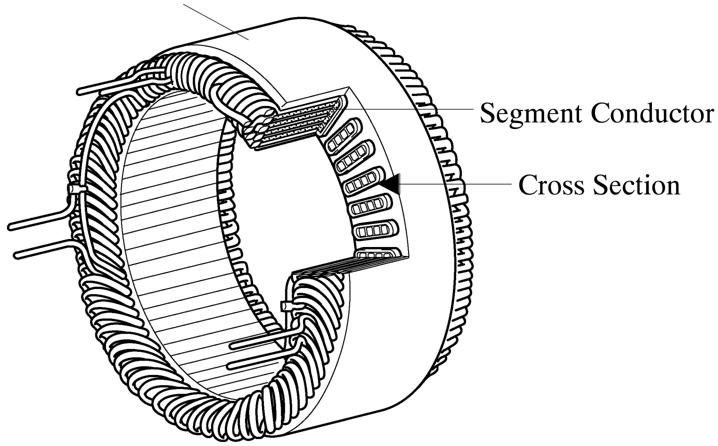
2. Construction and Operation

This generator uses a joined segment conductor system, in which multiple segment conductors are welded together at the stator. Compared to the conventional winding system, the electrical resistance is reduced due to the shape of the segment conductors, and their arrangement helps to make the generator more compact.



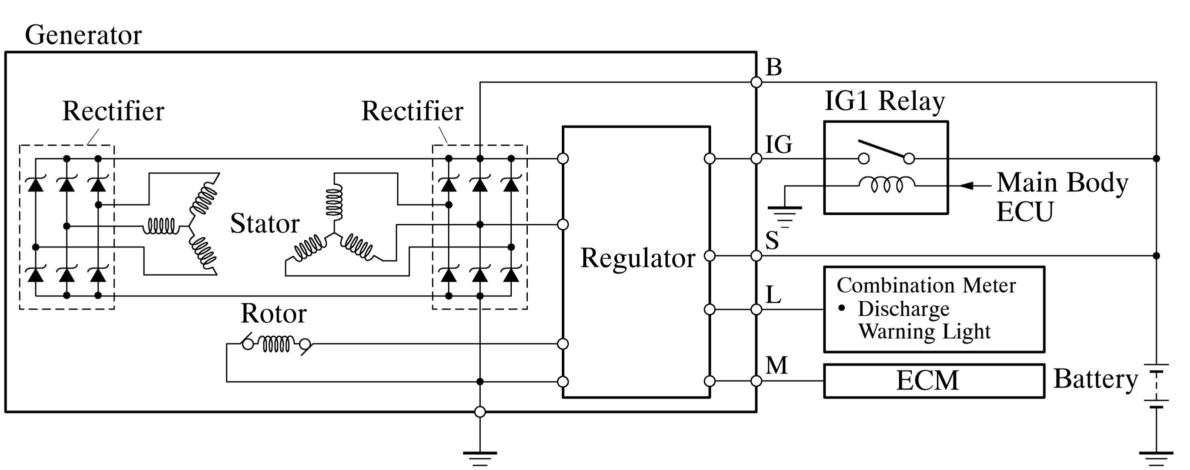
Segment Conductor Type generator 206EG40

Conventional Type Generator 206EG41



Stator of Segment Conductor Type Generator 206EG42

▶ Wiring Diagram ◀



080EG16S

3. Dual Winding System

A dual winding system is used. This system consists of two sets of three-phase windings whose phases are staggered 30°. This system results in the reduction of both electrical noise (ripple and spike), and magnetic noise (a hum that is heard as alternator load is increased). This system significantly suppresses noise at the source (alternator). Because the waves that the respective windings generate have opposite polarities, magnetic noise is reduced. The magnetic noise is significantly reduced, but the electrical power generated does not cancel itself out due to the use of separate rectifiers. The opposite polarities that are generated can be seen below.